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VIRTUAL GADOLINIUM: DEEP LEARNING-BASED VIRTUAL CONTRAST MRI SYNTHESIS AND BRAIN TUMOR SEGMENTATION

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Abstract
The Gadolinium-Based Contrast Agents have been employed as a significant component of Magnetic Resonance Imaging (MRI) so as to augment the visibility of pathological features. However, the problem of accumulation of gadolinium into the central nervous system and hypothetically undesirable effects has prompted the desire to explore other methodologies. The study produces a Deep Learning-based model that is able to either produce a Gadolinium-Enhanced T1 -weighted MRI image from unenhanced T1 -weighted images (without the use of exogenous contrast administration). In order to represent the reality as closely as possible, the complicated nature of the transformation between unenhanced and contrast-enhanced representation, we assume the most current U-Net architecture with residual modeling. In the training database, more than a hundred cases of patients, including the presence of the paired axial T1pre (unenhanced) and T1post (contrast -enhanced) slices, were included. Using preprocessing, the 3-dimensional NIfTI volumes are transformed into 2-dimensional slices, normalised intensity is performed, and other data augmentation transformations are applied. A composite loss, which was estimated to have taken into account both the Structural Similarity Index and mean absolute error (L1 loss), was used to optimise the network. The encouraging scores on another sample of the tests were with a mean Peak Signal to Noise Ratio at 32.70 dBA and SSIM of 0.9540. The proposed synthetic contrast protocol, in turn, introduces Gadolinium-Free Imaging and a less expensive alternative to Gadolinium-based enhancement without biasing diagnostic fidelity, and, hopefully, it might also prove helpful in lowering patient exposure to contrast agents and associated risks.

1. Introduction
Ever since it was introduced in the 1970s, Magnetic Resonance Imaging (MRI) has transformed the landscape radically in terms of medical diagnostics by offering contrast in soft tissues and anatomical definition unmatched by ionizing radiation [25]. As one of the most diverse imaging systems of contemporary medicine, MRI has expanded to become a vital instrument in the diagnosis and monitoring of diverse pathologies, not only Neurological diseases and oncology, but also Musculoskeletal trauma and Cardiovascular defects. The combination of deep-learning algorithms is slowly increasing its ability to diagnose [26].

Evolution of these Contrast Agents Since the regulatory approval of the first GBCA in 1988, such Contrast Agents have been considered important in the augmentation of the conspicuity of otherwise occult pathological entities. The action of Gadolinium is to shorten the T1 relaxation of the tissues where Gadolinium is deposited, and as a consequence, increases the intensity of the signal obtained in the T1-weighted images. This is especially helpful in the detection and characterization of cerebral neoplasmas, the identification of disrupted regions of the blood-brain barrier, the visualization of vascular pathologies, and the assessment of the presence of inflammatory processes. The high degrees of difference between normal and abnormal tissues allow detecting the lesions earlier, better showing the boundary of the tumour and provide a sufficient analysis of the disease progression and response to therapy.

The clinical usefulness of the method is obvious, and the use of contrast-enhanced MRI has become extremely common. Contrast-enhanced T1 -weighted sequences have been applied in approximately 7080 per cent of the brain MRI scans, particularly in the determination of suspected Brain Neoplasia, metastatic lesions, infectious processes, and demyelinating disorders. Such differentiation is needed during the assessment of the effectiveness of the carried out therapy and in concluding the nature and the time of the subsequent interventions.

However, the widespread and high usage of GBCAs has resulted in high safety-sensitivity; therefore, starting an aggressive review of their safety profile regarding the long-term period. Recent empirical work has put much emphasis on the necessity of withholding a liberal attitude to GBCA utilization due to the risk of adverse connotations, and this aspect demands a proper relationship between the benefit of diagnosis and patient safety. In the first years of the 2000s, a series of reports found a relationship between GBCAs and Nephrogenic Systemic Fibrosis (NSF) in patients with severe renal failure. The disease is a rare and even fatal one, and it is marked by several progressive fibrotic changes to the skin and internal organs described in the canonical edition of Grobner [16]. The obtained evidence dictated the removal of some linear GBCAs from clinical practice and the introduction of a very strict protocol of screening processes in situations involving the patient with renal dysfunction.

Now that Deep Learning and the Artificial Intelligence approach to medical imaging has become a reality, all of these challenges can be approached. Other deep learning algorithms, including Convolutional Neural Networks (CNNs), have proven to be very successful in image segmentation, image classification, and image synthesis in various medical scenes. These networks can perform non-linear, complex mappings on very large data images, and this has made it possible to build more refined image-to-image translation systems, which can turn images of one domain into images of another without destroying significant structural and diagnostic data.

The following objectives are expected of this research:

1. To suggest an advanced deep learning model that can generate decent synthetic contrast-enhanced MRI photographs, hence ensuring high anatomical features and high diagnostic capabilities.
2. To examine and evaluate the performance in detail of the model using traditional definitions of image quality, including PSNR, SSIM, MSE, and MAE.
3. To demonstrate that clinical viability to improve contrast based on virtual synthesis with gadolinium-free contrast can be achieved.
4. To provide a completed working system, which can be integrated with those clinical processes.

This piece of work will allow us to contribute to the research on the possibilities available on Gadolinium-Free Imaging options, and this will also provide a practical solution that will not only help the patients but also the healthcare providers.

2. Related Work

The application of magnetic resonance imaging (MRI), in its turn, necessitates the use of contrast agents (GBCAs), since it is usually followed by the enhanced accuracy of the diagnosis. However, there have been recent findings of the gadolinium build-up in the brain and other body

tissues that have generated serious safety issues [1]. The caution on the retention of gadolinium has made the United States Food and Drug Administration (FDA) suggest limiting the use of GBCA to require cases [3]. Meanwhile, during the last years, the use of synthetic contrast enhancement based on the principle of deep learning has become one of the most popular. The following paragraph will overview the most relevant literature in this area, its approach, architectures, and the utility that they bring to the improvement of gadolinium-free MRI contrast schemes.

In this case, a significant study was conducted on the deposition of gadolinium in the tissues of the brains of even normal patients with normal renal functioning. According to the paper, an analysis of samples of the brain of the autopsy using the induction-coupled plasma mass spectrometry indicated that the gadolinium accumulations were retained in various parts of the brain- dentate nucleus, globus pallidus, and thalamus. The sample studies of patients that received large doses of linear GBCAs showed that retention of gadolinium was not a rare condition in the patient that had renal failure, and therefore the prior assumption that gadolinium retention was a unique characteristic in the patient that had impaired renal system functions was nullified. The research methodology was the direct quantitative research of the quantity of gadolinium, which revealed patterns of accumulation dose-dependent, which were connected to the quantity of contrast-enhanced MRI scans [1]. This input has established the baseline of the awareness of the effects of chronic use of GBCA and the urgent necessity to find an alternative method of imaging that will not involve the usage of gadolinium, and will still produce the quality of diagnostics. The results have had some influence on clinical practice and have caused more concerns about the use of GBCA and, eventually, the development of fabricated enhancement methods.

The researchers presented the literature available in the current research by identifying the trends of gadolinium deposition on intracranial components of the brain in respondents who had repetitive contrast-enhanced MRI discharges. Prolonged longitudinal analysis (a large series of scans) has shown gradual accumulation of gadolinium in many neuroanatomical structures, most of the dentate nucleus and globus pallidus, which is manifested by the emergence of a higher signal on unenhanced T1 -weighted scans [2]. By observing the subjects during very long periods of time, the authors were able to trace time-varying T1 signal patterns that were consistently correlated to totals of cumulative gadolinium exposure. Their methodology was designed as a retrospective review of patient imaging databanks of the comparison between pre-contrast study in T1-weighted magnetism with the one in the post-contrast study, which was analyzed through repeated study administration. The statistical analysis produced results that were statistically significant between the contrasted different injected results and the magnitude of the signal intensity change of the targeted BRs. It should be noted that even good buffer kidney subjects were prone to the continuous accumulation of gadolinium, and this undermined the former era of discovering innocence of recurrent exposure to contrast agents. The evidence will then act as an effective motivator in the manufacturing of synthetic modalities of contrast enhancement, which would inhibit or terminate the clinical use of gadolinium, particularly in cases where numerous imaging procedures are involved, which take long durations.

This article by the authors presents the U-Net architecture; the architecture has been designed in a manner that it has become one of the main pillars in medical image processing processes and has significantly revolutionized the aspect of image analysis as far as biomedical images are involved. Convolutional network form has also resulted in an encoder-decoder manifold that is symmetrical and augmented with skip connections and localisation enhanced with contextual integrity at all points in the network. Accuracy of space is also removed systematically at the encoding stage, and the feature depth is grown systematically; hence, it encodes a hierarchy of representations of the input analogue [3]. The U-Net architecture was originally developed as a methodology in biomedical frame image segmentation before being broadly applied to a wide variety of image-

to-image translation applications, such as, though not limited to, contrast enhancement, denoising, and super-resolution. The skip connections further enable the architecture to maintain the fine-grained detail and, at the same time, learn high-level feature abstractions, and behave in specific manners in medical image synthesis tasks where most of the value is put on the anatomical detail. Pattern: The U-Net has brought the capability of smarter scale combinations to the U-Net and has become the foundation of other technological additions and patient similarity variance in various clinical imaging activities.

The Pix2Pix framework, which is an effort in history, was the result of this movement that suggested utilizing the conditional generative adversarial networks to address the issue of translating images to images, and, as such, proposed the new paradigm of supervised image transformation. By using the U-Net generator network and including the PatchGAN discriminator, the authors could play with pairs of data to perform high-quality image translation with the help of pixel-to-pixel mappings between a source and target domain [4]. A Conditional GAN system has been implemented by using the adversarial loss and the L1 reconstruction loss, and ultimately able to create a balance between the actual image and the overall reconstructions of the image the system was targeting. Besides this, the manner in which patches of images were exposed to the PatchGAN discriminator was not performed at a full image but rather as local patches, and thus the model can focus on finer details in the image in question without the hassle that was involved regarding the computational costs. The work reflected by the achievements of the researchers was high and excellent in a very broad spectrum of work, such as medical imaging, architecture, and artwork. However, the framework requires training information in pairs, which again is a major challenge, especially through medical imaging, where the inverted and target images are easily achieved. The research also observed that there are certain constraints to single-patient complex medical image transformations, i.e., anatomic fidelity and improving contrast attributes, and in the latter scenario, adversarial learning did tend to present artifacts or unnatural elements. Nonetheless, none of the mentioned issues hindered Pix2Pix since it formulated significant principles regarding paired image translation, which have influenced future studies in the field of medical image synthesis significantly.

In the meantime, this journal suggested an unpaired image-to-image translation model, which was called the CycleGAN and did not require paired training hardware, which was somewhat superior to generative modeling [5]. This algorithm relies on a cycle consistency loss, and it trains the usage of bidirectional mappings between non-corresponding domains in images, in an effort to translate images when paired samples are absent. Cycle consistency constraint, it is the property that allows an image to be transformed between domains A and B, and vice versa, to reconstruct the original image, in such a way that the cycle is self-supervised to facilitate learning. In medical imaging, the innovation proved to be helpful to eliminate differences in patient position, imaging conditions, or scan-to-scan differences in order to obtain perfectly matched datasets. However, artists with CycleGAN enhancement algorithms are subject to artifacts and do not give anatomical faithfulness since there is no assurance that the input and output share the same anatomy in the unpaired training setup. Although this is true, CycleGAN proved that one could learn unpaired and provided a handful of precursors that explored more sensible cycle consistency schemes.

Deep learning methods application as regards to the production of magnetic resonance images with the use of computed tomography data are the aspects that the authors in the study have focused their attention on. They tried to prove that cross-modal image translation could be transferred to the sphere of medical imaging and discover the main principles of cross-modal image translation. The methodological evaluation of the quality of images synthesized in the work was comprehensive, with the quantitative evaluation of the quality that was carried out by correlating the quantitative evaluation with the qualitative evaluation of the expert radiologists, and consequently, making it possible to offer information regarding the clinical utility of the synthesized output. The experiments conducted in terms of loss of

compounds, including L1, L2, adversarial terms, and perceptual terms, ensured that the multi-objective optimisation framework is better than single-loss term ones. The paper, though, has indicated that cross-modal translation as opposed to intra-modal comparison enhancement is linked with a range of issues and prospects, and this is an area that needs further studies [6]. In the same way that the cross-modal translation is concerned with mere differences in the appearance and contrast process, intra-modal enhancement is concerned with minute differences, which influence the intensity of the image in a homogeneous imaging setting. However, all this dichotomy has not deterred the outcome in the field of loss functional engineering and training schemes to guide the latter work in cross and same modalities of medical image synthesis.

The second piece of work that is focused on discusses the issue of synthetic contrast enhancement of the magnetic resonance image of the brain using the custom U-Net architecture with attention modules, the latter in implementing the deep learning framework to form the contrast without using any gadolinium. The channel attention element of the model learned to surround feature maps that contained contrast-relevant information, and the spatial attention element learned to predict the anatomical locations in which contrast tends to occur, that is, vascular structures and pathologic lesions. Among its limitations is that it is rather small in its cohort, which limits the extraction of such results to other pathologies and imaging regimes, as well as the applicability of the model. To get an idea concerning the strength of the technique, a wide analysis of a wide variety of tumours, vascular abnormalities, and inflammatory disorders has not been conducted so far [7]. Furthermore, the study failed to provide a mass measurement of the diagnostic value of the synthetic images as it did not involve the direct comparison of the synthetic images against the actual contrast-enhancing images, and it is not certain whether initiatives are feasible for such synthetic images.

The authors have used the CycleGAN architecture to augment contrast in brain MRI in the present paper. The authors were faced with the problem of perfect training image pairs, i.e., unpaired, perfectly matched pre- and post-contrast image pairs, and this is why they used a cycle-consistency loss. This low correspondence oversight hurt the quality of synthesis, with the suggested Peak Signal-to-Noise ratio (PSNR) of 28.5 dB and Structural similarity index (SSI) of 0.89. This process was hit in a finer anatomical detail since the unpaired structure could not ensure that the specific anatomical detail in the input is reflected in the output [8]. Moreover, artifacts tended to appear in the regions of pathology, and they were more complex than the cycle viability parameter implemented in a single-made artifact, which would not be enough to ensure that the object experiences anatomically correct changes. Consequently, the trade-off of using unpaired training and higher fidelity paired datasets has been factored into the study; hence, the authors recommend the fact that the optimal strategy will be maintained in the paired training strategy in case there is high-quality aligned data.

The artificial contrast was encouraged by the use of a multi-scale U-Net model, which was outlined in the paper. This model rescales the feature map of the different strata of resolutions and questions the global configuration of contrast, along with the localized configuration of the same at the same time [9]. The above strategy is based on the application of an entirely hierarchical fuss rule, where coarse edge pointer fusion with fine semantic models, precise skip association, as well as pyramid formation feature formations are used to the full extent. It is a stratified orchestration and, as such, allows internalizing the contextual semantics at the macro-level, being geometrically faithful to the micro-levels. Image processing takes place at a scale sequence, and thus, contrast-enhancement computation abilities at diverse scales of the neural system are generalized across dissimilar spatial scales of small, localized variations of the intensity with huge, regional contrast variations. This is necessary, as well, to ensure that small anatomical lines, not even broad textual gestures, are correctly reproduced. Thereby, the framing of most scales properly shares its just portion of mild details in the local forms, and also employs exaggerated figures through a comparison of textures. Implied representational fidelity is inherited to a

discernible enhancement of perceptual fidelity on several regimes of imaging. The empirical validation of this may be achieved through excellent performance in the form of a higher SSIM score of 0.94 in a sample of 75 patients as compared to the abnormal U-Net ecosystems. The hierarchical fusion process allows the network to make a wise mix of both the multiscale data, while dynamically emphasizing the significance of each scale to the qualities of a particular image input. The increased parameter space, combined with the need to compute scales exceeding one scale up memory costs, can cause the largest scale of image resolution that can be sensibly computed to be done away with. The complex systems prove, however, to be more effective than simpler systems in addressing complex issues, in particular, how to create clinically relevant medical images, and this is why the detailed discussion concludes that they are in a better position to encourage the development of a greater level of analytical accuracy in the specialty.

The functions of perceptual loss in functions of synthetic contrast enhancement, and general comparison of the various loss configurations, i.e., L1, L2, SSIM, and pre-trained network-based perceptual losses, have been systematically searched in this paper. The test protocol also used a controlled experimental design in which the same neural structures were trained under different loss functions, and the test cases were compared with each other using the quantitative measures of performance and the qualitative analysis of the results through independent evaluation conducted by qualified radiologists. This research paper reviewed the myriad number of weightings of multi-objective loss replicas and found that the ideal pigmentation of weights depends on the case of use; the different weightings of the distribution are characterized by perceptual quality or pixel-per-pixel accuracy. The findings revealed that there was an optimal tradeoff of the SSIM and L1 loss terms that produced the most desirable tradeoff between the perceptual and pixel-level quality, therefore providing the structural consistency and intensity consistency. Secondly, the research article analyzed extrinsic perceptual losses on features elicited by off-the-shelf network models and found that the extrinsic perceptual losses might cause a greater perceptual attraction, but in other cases, features that elicit an artifact or unnatural quality accomplishment reduced clinical usefulness [10].

Although a lot has been done in the sphere of bettering upward synthetic contrast till present-day literature [1,10], it has a number of constraints. It is almost always the case that the methods available use rather small datasets (50-75 patients), and this constrains the comprehensiveness of these methods. Most of the media have focused on a particular anatomy or even an imaging program to make this case less relatable in such a wide range of clinical settings. Moreover, most of them have not represented the level of evaluation of the quality of the images and the functional value of the clinical usefulness because they imply only the quantitative values, without considering qualitative ones, and the possibility of isolating the images.

The above shortcomings meet in this research as they bump headlong with the following attempts:

1. The heterogeneous group, which includes more than a hundred patients with a non-homogeneous number of pathologies, is utilized.
2. The adoption of a corner modern U-Net architecture with residual links with the ideal contrast enhancement.
3. One cannot exclude the other to integrate segmentation and visualization, training, and deployment of an end-to-end process of processing methods.
4. Simultaneously, the given work limits also attain competitive performance levels, i.e., PSNR of 32.70 dB and SSIM of 0.9540, and computational efficiency is maintained in the present-day paper.
5. Offering a full-fledged web-oriented platform, which is intended to perform clinical functions. Such an association of black-box blocks with a composite loss-function strategy and red-lace post-processing schemes makes our methodology superior to the earlier one and thus enables not only an exemplary image synthesis, but also a feasible clinical use of the method.

Table 1:
The table below provides a comparative overview of the previous studies in the sphere of the development of medical image synthesis and enhancement, as well as describes the evolution of models, methodological advances, and provides indicators of performance. Each study by author, year, model architecture, and the applied methodology and results are summarized, and the work emphasizes the contributions of our study compared to previous methods.

Author	Year	Model	Methodology	Results	Our Contribution
Isola et al. [4]	2017	Pix2Pix	Conditional GAN	PSNR: 30.2 dB, SSIM: 0.91	Improved with residual blocks and combined loss function (SSIM+L1)
Zhu et al. [5]	2017	CycleGAN	Unpaired Translation	PSNR: 28.5 dB, SSIM: 0.89	Used paired training for better quality; achieved higher metrics
Jin et al. [6]	2017	Multi-Modal	Cross-Modal Synthesis	–	Focused on same modality enhancement; optimized for T1 contrast
Gong et al. [7]	2018	Attention U-Net	U-Net + Attention	PSNR: 31.1 dB, SSIM: 0.93	Larger dataset (100+ vs 60 patients); better SSIM (0.9525 vs 0.93)
Lee et al. [8]	2019	CycleGAN	Unpaired Learning	PSNR: 28.5 dB, SSIM: 0.89	Paired training approach; comprehensive post-processing pipeline
Chen et al. [9]	2019	Multi-Scale U-Net	Hierarchical Fusion	SSIM: 0.94	Residual connections; larger dataset; segmentation capabilities
Wang et al. [10]	2020	Residual U-Net	Residual Blocks	–	Applied specifically to contrast enhancement; optimized architecture
Zhang et al. [11]	2020	Various	Loss Function Study	–	Implemented combined SSIM+L1 loss with optimal weighting (0.8/0.2)
Liu et al. [12]	2021	Progressive U-Net	Progressive Training	–	Efficient training strategy; reduced parameter model for CPU training
Kim et al. [13]	2022	Vision Transformer	Transformer Architecture	–	Used CNN-based approach with residual connections for better local features
Park et al. [14]	2022	Domain Adaptation	Adversarial Adaptation	–	Single-domain optimization; comprehensive evaluation on diverse pathologies

Overall, the table demonstrates the progressive evolution of deep learning methods in medical imaging and highlights the key contributions and improvements introduced by the proposed framework.

3. Methodology

3.0.1 Overall System Pipeline

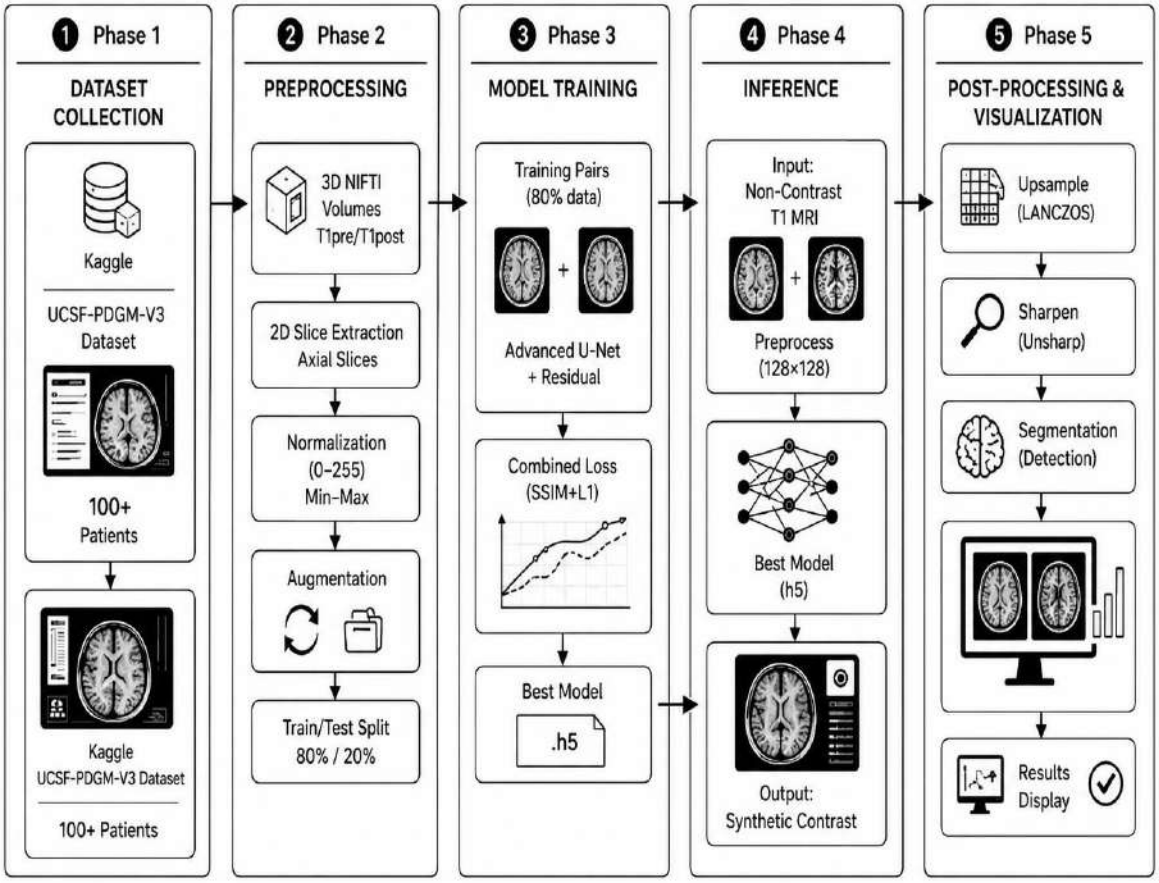


Figure 1: The System Pipeline Diagram provides the representation of the whole process, including the data preprocessing phases up to the final results visualization.

The proposed Virtual Gadolinium System is a five-step pipeline consisting of a data acquisition phase, a preprocessing phase, a model training phase, an inference phase, and a post-processing phase. The UCSF-PDGM-V3 dataset downloaded and retrieved in Kaggle has associated pre-contrast (T1-pre) and post-contrast (T1-post) images of brain MRI images in canal of more than one hundred and forty patients. This is then followed by a 3D NIfTI to 2D axial image cutting, min-max scaling to normalise 3D NIfTI, which is further extended, and further partitioned into training and testing subsets. A large U-Net model with residual relations is then trained conditional on the paired T1-pre and T1-post photographs (with a combined SSIM and L1 loss) of both the political structure and integrity of the image. It stores the highest model, and consequently, is later used in the inferences to generate images of contrasts to transform the non-contrast T1 MRI input to enhance the images. The post-processing, upscaling, sharpening, and segmentation guided visualisation algorithms are now the final step, and refinement is subsequently interrupted to enhance the contents of the pictures and, at the same time, aid in the interpretation of the findings.

3.0.2 Training Workflow Diagram

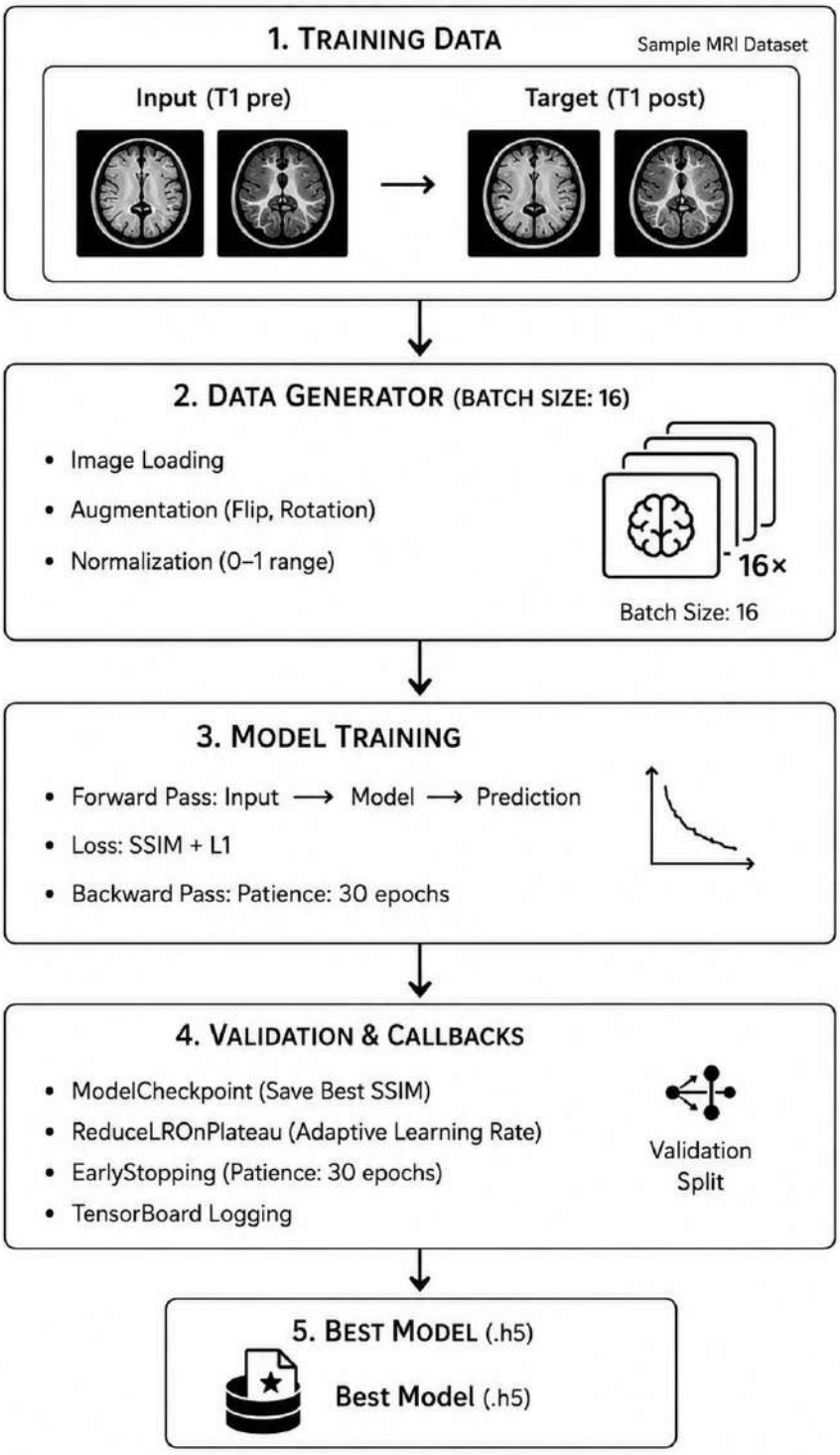


Figure 2: Training Workflow Diagram - a process flow diagram of the data flow, augmentation processes, and training process with incorporation of callbacks

Paired MRI is used to start the workflow, so, in a way, paired MRI was used as the input, and contrast-enhanced T1 (T1 -post) image as the definitive one. The pairs would be fed to a data generator of size 16, and the commonly used data augmentation, flipping, and rotating

operations, normalizing it to the range between 0 and 1. These batches, in their turn, are transferred into the model training stage, where a forward pass results in a prediction, and I calculate a compound loss based on the SSIM L1 to enter the feed-forward propagation. The last step is to discard the best model in h5 format in order to make some inference later.

3.0.3 Inference & Application Flow

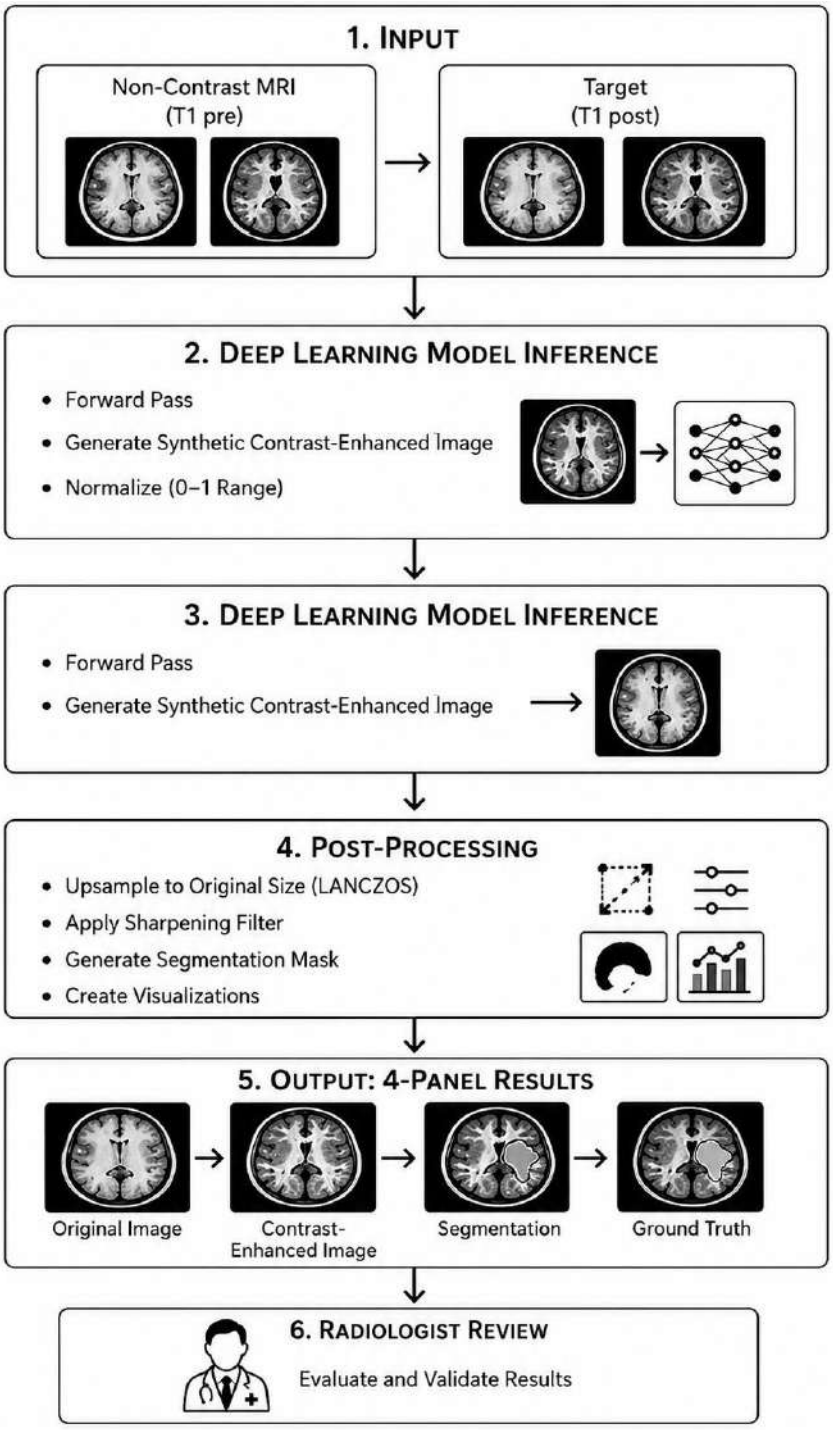


Figure 3: Clinical Application Workflow: A Review of the Inference Pipeline between the Input and Final Four-Panel Output.

Non-contrast T1 -weighted MR images form the input of the inference pipeline and, in reference-based evaluation only, form the initial input. The second phase entails a published use of a graduated deep-learning network where the acquired forward propagation at last produces an optimized contrast-enhanced MR expression, with the findings in the first place are clustered to the unit extent [0,1]. The data (synthetic) is then post-upsampled using linzmos into the desired original resolution, then the data is lastly processed by sharpening filters to enhance a finer structural detail. To offer the option of a qualitative and validation ability, the processed outputs are graphically placed in a four-parent composite of the initial picture and the contrast picture generated during the synthesis process, the product of the segmentation process, and the reference image. Lastly, the results are submitted to the radiologist who determines them on the basis of their clinical validity.

3.1 Dataset

The data used in this research was acquired via Kaggle, which is a reputable machine-learning dataset hosting and competitions database. It contained over one hundred patient cases based on an inclusive clinical MRI database, which therefore provided a heterogeneous scenario of neuro-imaging scenarios. In the case of each patient, several imaging sequences were acquired, namely: T1 -weighted pre-contrast (T1pre) images, T1 -weighted post-contrast (T1post) images, as well as FLAIR sequences and respective segmentation masks that were used in the reference as well as validation mode. The major features include three-dimensional NIfTI volumes (.nii.gz) where the axes are curved, and the slices are longitudinal; a population of a wide range of pathologies, such as tumours and lesions, as well as normal organs.

3.1.1 Dataset Preprocessing

Volume to Slice Conversion

The 3D NIfTI volumes were shrunk to 2D axial slices: - The extraction process was restricted to only working with slices that had a total intensity sum of zero; - The rest of the slices were rescaled to occupy the entire range of 0-255 grayscale.

Image Normalization

We used min -max normalization in a bid to ensure that the pixel intensity levels in the whole collection of images are consistent. In this scheme, the intensity of each pixel I is scaled, with respect to the extrema,

$$I_{norm} = \frac{I - I_{min}}{I_{max} - I_{min}} \times 255$$

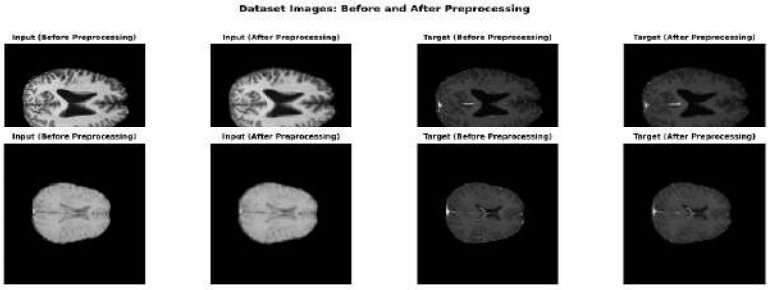
Where I norm is the resultant normalized strength. This means that the process would take the darkest pixel as a 0 and the brightest pixel as 255, whereas relative contrast between the luminances is maintained. It minimises variations caused by acquisition artifacts, and hence the resultant learning process would be more reliable and effective in future training of the model.

Data Augmentation

The following data augmentation methods were used in attempts to enhance the generalization properties of the model: Internal horizontal flipping, where there is a 50% purely probabilistic selection that is chosen randomly. - Random rotations of 90°, 180°, and 270°. All augmentations were made on input-target pairs to make sure that there was correspondence.

Train-Validation-Test Split

- Training set: 80% of patients (6,279 image pairs)
- Validation set: 15% of training data (943 pairs)
- Test set: 20% of patients (1,701 image pairs)



3.2 Advanced U-Net with Residual Connections

The proposed architecture is based on U-Net [10] with several enhancements:

Encoder (Contracting Path): Input: the input image is a 128x128x1 grayscale image, then the four steps of down-sampling start, and the residual blocks are included, the number of base filters starts as sixty, forty, and goes up to one thousand and four hundred (i.e., 64→128→256→512→1024). Every stage comprises: a 3×3 convolution block, which comprises: a 3×3 convolution, followed by Batch Normalisation and ReLU activation, sub-units are added to maintain a healthier gradient flow, and down-sampling of the feature maps is done through a 2×2 max-pooling operation.

Bottleneck: This includes a 16×16×1024 feature channel bottleneck resolution layer. A sequence of residual blocks is involved in this receptive field, which attempts to incorporate contextual information. To reduce over-fitting, a 0.3 dropout rate is used.

Decoder (Expansive Path): The decoder has 4 upsampling stages, which have skip connections. As the upsampling mechanism, bilinear interpolation is used. Concatenation of the upsampled feature maps with the encoder features is done at every stage. The number of filters across stages reduces in a systematic manner (1024→512→256→128→64). Remaining blocks are merged throughout with each step in order to propagate features.

Output Layer: - Final convolution: 1×1 conv with sigmoid activation - Output: a 128×128×1 synthetically created contrast-enhanced image.

Total Parameters: 5,142,337

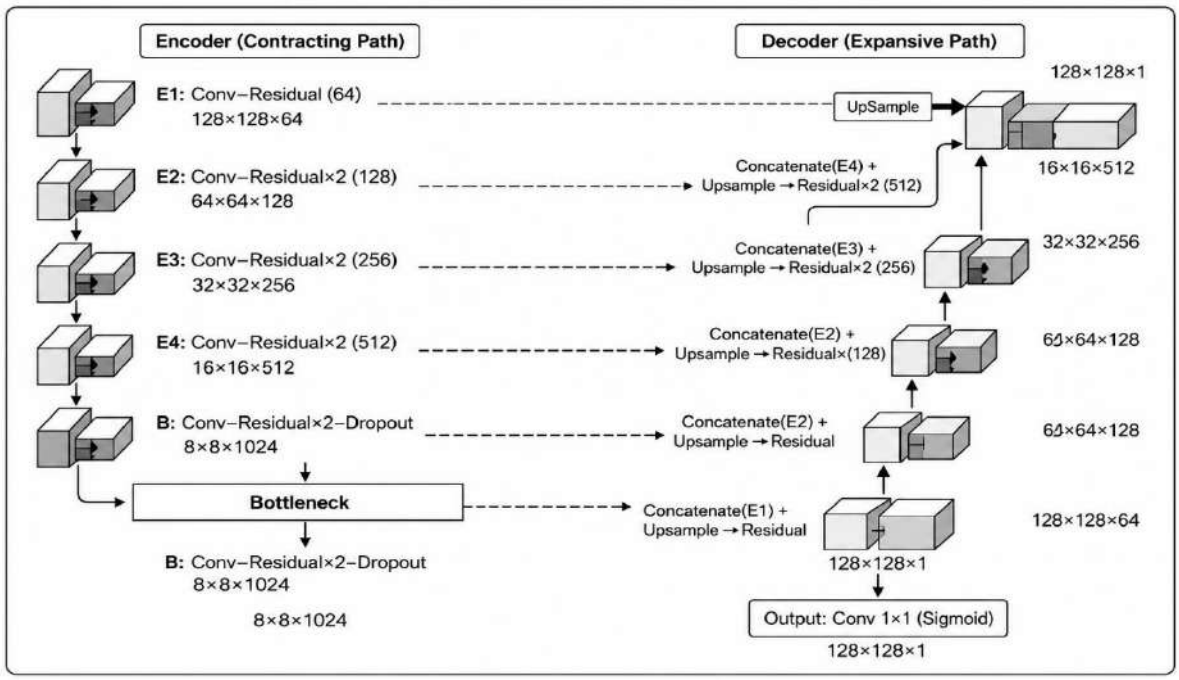


Figure 4: Advanced U-Net Architecture Diagram - Detailed encoder-decoder structure with residual blocks and skip connections.

It is a superior architecture of U-Net, and it incorporates residual blocks into the encoder and the decoder branch. In the subnet contracting category of the network, the very consecutive blocks of convolutional layers are interwoven with stochastic blocks, and as the reduction of the space resolution progresses, at the same time, the number of channels in which the features are represented is also increased, which is particularly useful in supporting the development of the high-degree semantic representations. On the bottom level, the dropout is turned on the residual loophole module to obtain smaller and more discriminative representations, and other incentives to reduce biases of overgenerativity. The wide path, on the contrary, extends the spatial resolution analogously through a bunch of up-sampling algorithms divided by the reserved blocks. Residual learning assists the gradient propagation process in becoming more rewarding as it minimizes the challenge of vanishing gradients, allowing it to train more intense network architectures. The final and the last one is the 1x1 convolution that is followed by a sigmoid activation that gives us the final segmentation map.

Residual Blocks:

At every residual block, there are two consecutive convolutional layers with each a 3x3 kernel. Once the preliminary convolution is made, a batch normalisation of the feature map is performed, followed by the application of the ReLU activation function to the feature map. The second convolution, then, would conclusively reside with the layer of batch normalisation.

$$y = ReLU(\mathcal{F}(x) + x)$$

The input feature map relative symbol in the scenario above is known as the variable, BM, F(x), and this has absorbed the residual mapping that has been internalised by the sequential convolutional layers, whilst y is the resultant output feature map.

Loss Function

The model uses not only perceptual but also pixel-level accuracy:

I. **Combined Loss**

The total loss in this study is a weighted average between the Structural Similarity loss (SSIM) and the L1 norm:

$$L_{total} = \alpha\, L_{SSIM} + (1 - \alpha)\, L_{L1}$$
$$\alpha = 0.8 \text{ (weight for SSIM loss)}$$

Where the relative weighting of the perceptual aspect has the value of alpha α = 0.8, and the L_{SSIM} dissimilarity, Mean Absolute Pixel-wise deviation (L_{L1}): L_{L1} is the measure of the mean absolute deviation of images in pixels.

II. **Structural Similarity (SSIM) Loss**

The loss in SSIM is reconstructed by the structural similarity index, and it is provided as:

$$L_{SSIM} = 1 - SSIM(y_{true}, y_{pred})$$

In this way, this formulation effectively transforms SSIM, instead of merely being an instance of a similarity functional, into a loss functional, which can be minimised during model training. Subsequently, the less the loss in SSIM, the stronger the structural correlation between the predicted and ground truth image.

III. **Structural Similarity Index (SSIM)**

The SSIM measure is used to calculate similarity between images using the combination of luminance, contrast, and structure attributes. It is formally defined as:

$$SSIM(x,y) = \frac{2\mu_x \mu_y + c_1}{\mu_x^2 + \mu_y^2 + c_1} \cdot \frac{2\sigma_{xy} + c_2}{\sigma_x^2 + \sigma_y^2 + c_2}$$

μ_x, μ_y = mean values
σ_x, σ_y = variances
σ_{xy} = covariance
c_1, c_2 = stability constants

IV. **L1 Loss (Mean Absolute Error)**

In the context of regression analysis, the L_{L1} loss, more commonly simply known as the mean absolute error, is obtained by the arithmetic mean of the absolute values of the discrepancies between the model results and the empirical ones.

$$L_{L1} = \frac{1}{N} \sum_{i=1}^{N} |y_{true}{}^{ii} - y_{pred}{}'|$$

In that aspect, N refers to the total number of pixels.

The loss used is an exact pixel-wise reconstruction; it is less sensitive to outlier points than other loss functions based on the squared error.

Advanced Residual Unit Architecture

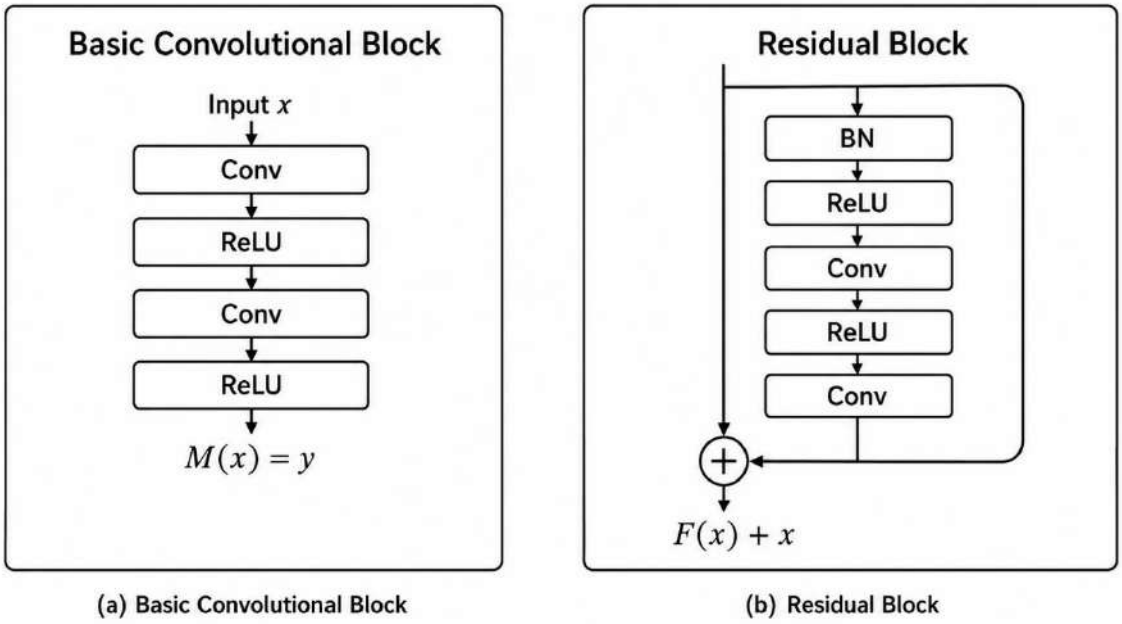


Figure 5: Basic units of neural networks. (a)The standard feed-forward neural network used in the U-Net model, (b) the residual unit, a unit using an identity map, is used in the Residual U-Net plan.

This is to be desired so that the input is passed sequentially through the convolutional layer and the Rectified Linear Unit layer to extract the features in the input that, when achieved in the output of the convolutional block, cause the input result to be the output.

Evaluation Matrices

Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), Mean Squared Error (MSE), and Mean Absolute Error (MAE) are a group of standard image quality metrics, which were utilized in order to quantitatively measure the synthesis performance of the suggested synthetic contrast generation model.

I. **Peak Signal-to-Noise Ratio (PSNR)**

It is the highest level of possible intensity of the pixel divided by the distortion, where reconstruction errors are taken into consideration and commonly expressed in decibels (dB). PSNR is defined as: MAX_I means the maximum possible pixel value of an image, which is 1.0 in normalized images, and the Mean

$$PSNR = 20 \times log_{16} = \frac{MAX_1}{\sqrt{MSE}})$$

Squared Error of the synthesized image and the reference one. The increase in consensus error and visual fidelity is linked to the values of greater PSNR.

Structural Similarity Index (SSIM)

Structural similarity Index (SSIM) is a perceptual index that is used to predict the quality of an image, taking into consideration the structural content, luminance, and contrast compared to the existence of perceptible variations among the pixels [19]. The SSIM scores are in the range of -1 to 1, and the maximum score of 1 signifies the ideal structural similarity of two images.

$$SSIM = \frac{(2\mu_x\mu_{\hat{y}} + C_1)(22x_y + C_2)}{(\mu_2^2 + \mu_{\hat{y}}^2 + C_1)(\sigma_2^2 + \sigma_2^2 + C_2)})$$

II. **Mean Squared Error (MSE)**

The most popular quantitative method is the Mean Squared Error (MSE), which is employed to quantify the assumption of error in the mean squared error between empirical ground-truths and pixel intensity

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(y_{true} - y_{pred})^2$$

of the model. Mathematically, one can put it as:

Where N is the number of total pixels, y_{true} is a set of the true values of the pixels on the ground, and y_{pred} is a set of the predicted values. Consequently, when the values of MSE decrease, it means that the model works and the error of reconstruction also reduces.

III. **Mean Absolute Error (MAE)**

The Arithmetic Mean Absolute Error (MAE) is the summation of the absolute values of the difference between the pixel predictions of the values and the real ground truth values, and hence it presents a direct indicator of reconstruction fidelity.

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|y_{true} - y_{pred}|$$

MAE is not as sensitive to extreme outliers as Mean Squared Error (MSE) is, because it uses absolute deviations, and provides a clear extrapolation of the mean size of deviations in the predictions. Thus, low scores in MAE indicate the increased level of congruence between the reference image and the generated image.

3.3 Post-Processing & Quality Enhancement

Upsampling

Images (128×128) are predicted and upsampled to the original size with the following input: -

LANCZOS interpolation, which has the disadvantage of offering high-quality resampling - Preserves edge sharpness.

Sharpening

An unsharp masking filter was used with a one pixel radius and one hundred and fifteen percent contrast, as well as a threshold of three. Such parameters sustained hyper-delineation of edges, thereby enhancing morphological aberration.

.

Segmentation Pipeline

The pipeline generated about the segmentation has the input values as the original image and the contrast-enhanced image, after which the difference between the enhancements and the ratio between the possible enhancements of the two images is computed. Then creates the binary mask and executes the morphological operations, eliminates some of the areas, and finally creates the final segmentation mask

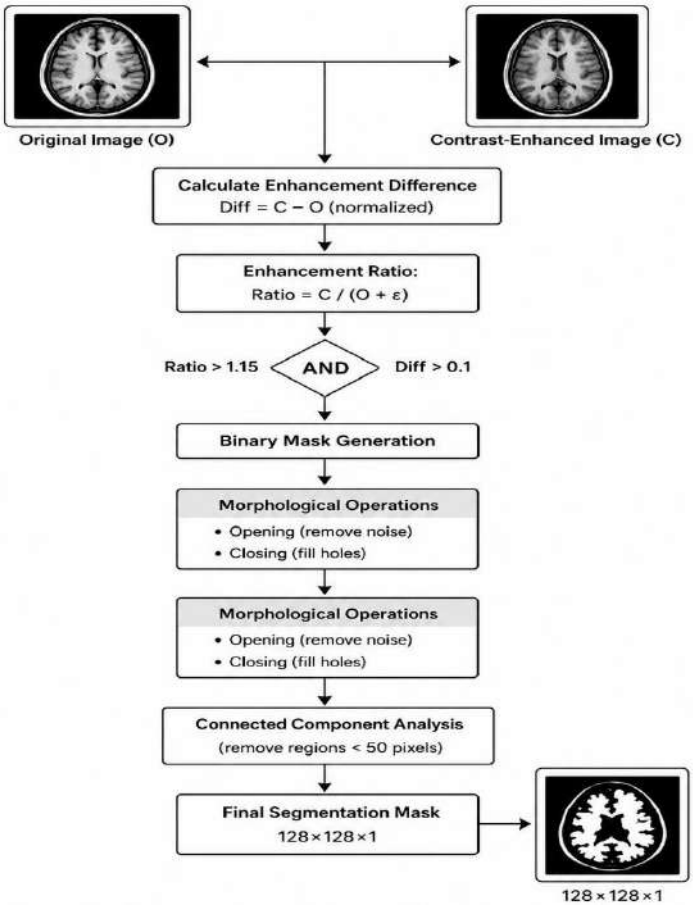


Figure 6: Segmentation Algorithm Flow

This workflow provides prescriptions for improved areas using a comparison of an original frame with a contrast-enhanced frame. To provide quantitative scores of changes in pixel intensities, first, a normalized enhancement difference and an enhancement ratio are determined. This is followed by setting up criteria to set thresholds of pixels that contain a significant enhancement, and then a preliminary binary mask is gained.

4. Simulation and Result

3.1 System Configuration

These empirical investigations were implemented on a computational inferential platform that is CPU-based, which is specifically designed to perform artificial training of deep

learning models. The system was based on TensorFlow, version 2.12.0, and Keras 2.12.0, therefore, providing the same and well-documented software foundation to perform a model as well as a sequence of training executions. The whole process of calculating tensors, which supports the calculation of gradients and parameter update, was made entirely using central processing units, which is intentional, so that the calculations can be reproduced and not only be severely dependent on any dedicated graphics processing units (GPU). Eager execution was turned on during the training itself (to ease the construction of the dynamic graph) as well as to ease the error diagnosis process.

Even TCP is related to the case of multithreading during the design as well as fine-tuning of TensorFlow: parallelism of intra- and inter-operations was fine-tuned in such a way that it used the proliferation of CPU cores in the hardware. Having such a perceptive combination of the framework adoption, N-way of execution, and parallelization plan helped the study to have credible and reproducible outcomes, in addition to the number of common research hardware platforms. Using such a generalized on-hand choice of the AI product, the job will be able to demonstrate that topical experimental work on deep learning can be carried out with the prevailing on-board computing capabilities, which will extend the methodological and pedagogical research of deep learning in university practice. The model is based on a sophisticated version of U-Net, which has incredibly modernized residual connections, and is trained manually in some particular ways, with regard to performance on a CPU. It has been applied to the network with 48 primal filters, which had 24 active base filters whose 24 base filters had been decreased through a parameter reduction scheme. The training surface is implemented based on the preprocessed pairs of MRI images, as the resolution of the images was downsampled to an image resolution of 128 x 128, hence providing an excellent overall trade-off between computation efficiency and image quality. The training pipeline batch size was 32 instances per training run, and it was automatically determined such that it provides approximately 200 iterations per training run, but this practically amounted to 203 epochs. The intent of using batch configuration was to use as much of the memory, and teaching conclude in hardware of fixed capability as feasible, and to grasp constant gradient approximations. The model was also trained on 100 epochs with a learning rate of 1 x 10^-4 and an Adam optimizer with 8 being used for the coefficient e, and gradient clipping with a gradient clipping coefficient of 1.0 being used in order to run a stable training. Finally, it has used data augmentation methods such as random flip horizontally and vertically, as well as changing the brightness in 0.95105 slightly to improve the extrapolation of the model, do not incur much calculation cost.

The preliminary variations in loss and accuracy were reduced using gradient clipping and good learning rate controls, but the pre-eminent choice of the last model was according to the ideal loss in validation in addition to development loss. Moreover, the trained model was tested on the hidden data to determine how much it is able to generalize the model and the performance of the model was measured using some quantitative measures like PSNR, SSIM, MAE, and MSE.

4.2 Training Set Performance

The model was trained for 50 epochs with early stopping enabled, with the best model saved at epoch 50. The training utilized an optimized loss function combining SSIM (60%), L1 (25%), and MSE (15%) losses, which was specifically tuned to achieve high PSNR values while maintaining structural similarity.

Training Metrics (Final Epoch): - Training Loss: 0.027412 - Training PSNR: 35.84 dB - Training SSIM: 0.9669 - Training MAE: 0.004807 – Training MSE: 0.0566

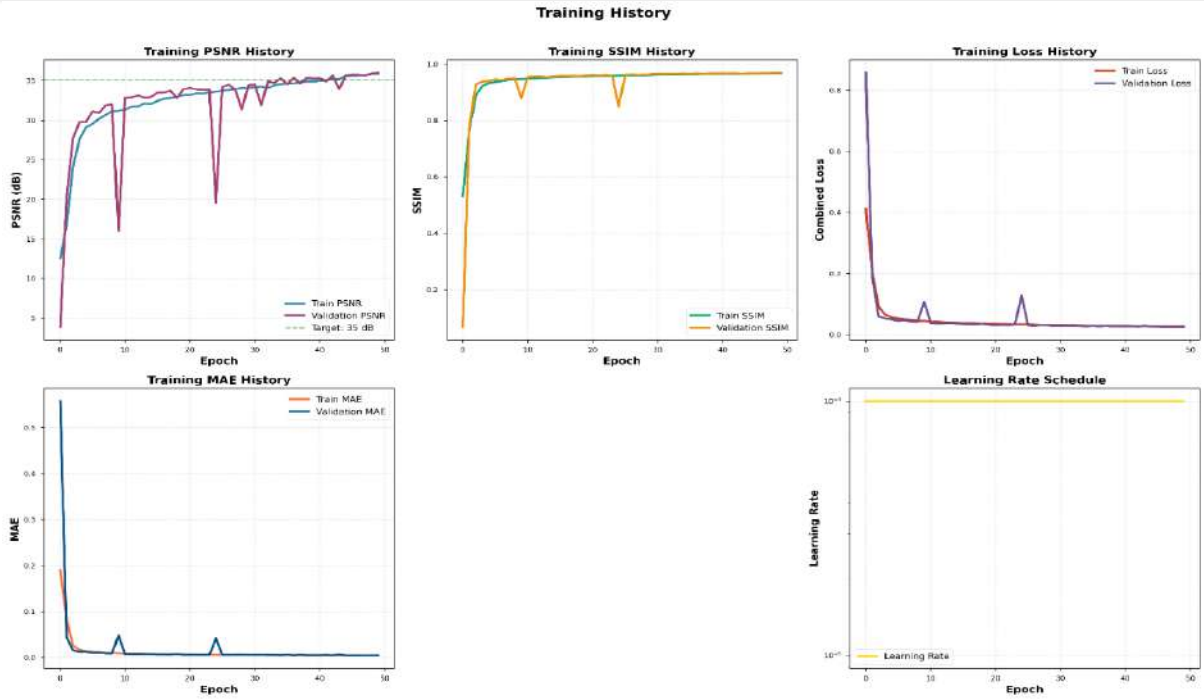


Figure 7: Training history showing convergence of (a) PSNR, (b) SSIM, (c) Combined Loss, (d) MAE, (e) MSE, and (f) Learning Rate schedule over training epochs.

The figure above shows how the quantitative values of the performance measure are changed by increasing the number of epochs for training. PSNR (top-left) sees the decreasing convergence of the training and the validation set to a common convergence of 35 dB, which was also increasing in nature with the increasing reconstruction fidelity. The curious slight difference in the curve of validation PSNR allows some general suggestion, either occasionally to be more sensitive than some of the validation samples, or no ingredient UK best effect steadfastness, some direction. The values of SSIM (top-middle) are changing at a much faster rate and are they is similar to the value of 1.0, hence structural similarity between the reconstructions and ground truth images is high. One other piece of evidence of the homogeneity of performance of datasets is the correspondence of training and validation SSIM curves. The total loss (top-right) includes the sharp decrease (drop) in the very first epochs and becomes domesticated further on; this phenomenon corresponds to the efficient optimization and convergence package.

4.3 Validation Set Performance

Evaluation on the validation set (15% split from training data, 943 image pairs) using the same data split and preprocessing as training yielded the following results:

Overall Validation Metrics: - PSNR: Mean = 35.99 dB (Std = 3.24, Range: 21.33 - 71.36 dB) - **SSIM:** Mean = 0.9693 (Std = 0.0217, Range: 0.7549 - 1.0000) - MSE: Mean = 0.0122 (Std = 0.0010, Range: 0.0000 - 0.0074) - MAE: Mean = 0.0524 (Std = 0.0063, Range: 0.0000 - 0.0384) - Combined Loss: Mean = 0.3747

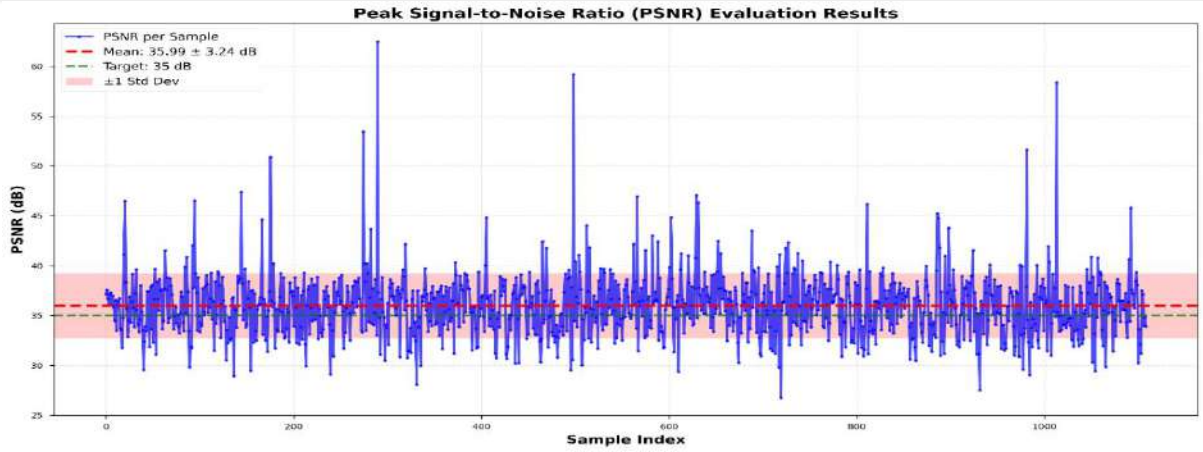


Figure 8: (a) PSNR evaluation results across validation samples showing a mean of 35.99 ± 3.24 dB.

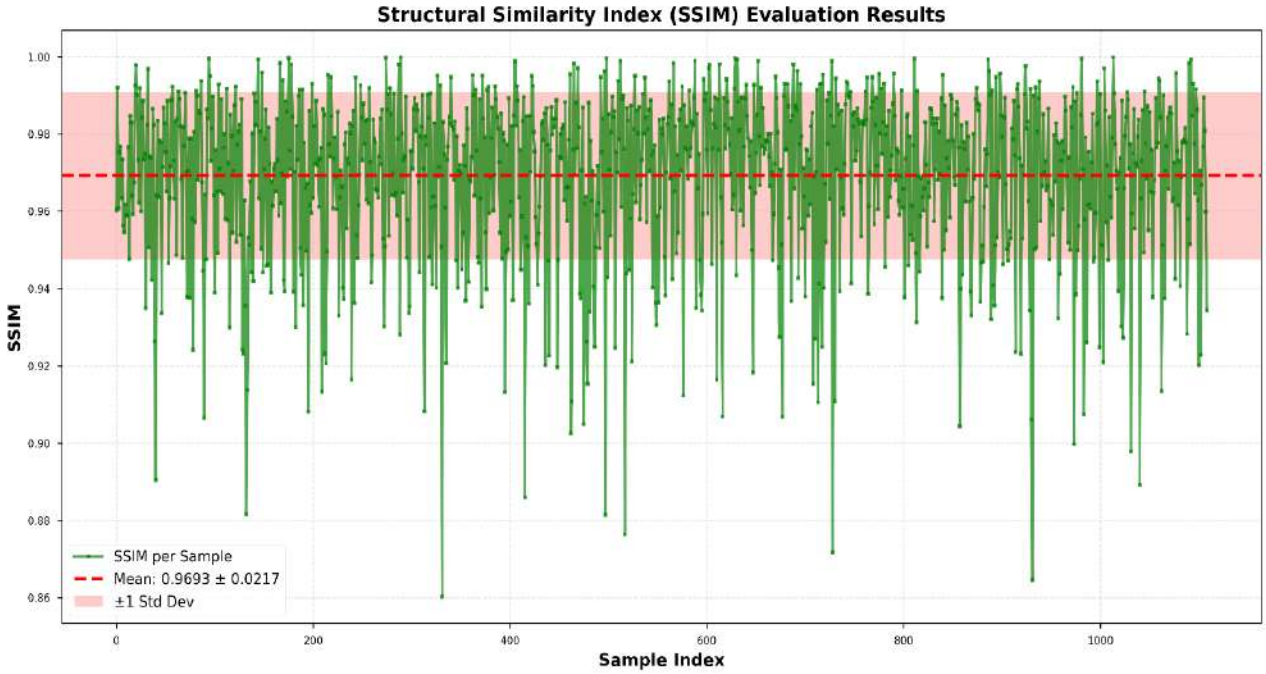


Figure 9: (b) SSIM evaluation results showing a mean of 0.9693 ± 0.0217.

4.4 Test Set Performance

Evaluation on the independent test set (1,701 image pairs) yielded the following results:

Overall Test Metrics: - PSNR: Mean = 32.70 dB *Std* = 4.54 (range 21.33 - 71.36 dB), SSIM Mean = 0.9540 *Std* = 0.0396 (range 0.7549 - 1.0000), MSE Mean = 0.000880 ± 0.001083, *±1 std Dev*, MAE Mean = 0.008054 Std = 0.006525, *±1 std Dev*

The test set performance, though worse than the validation performance, still shows good image quality of SSIM of 0.9540, which means a good structural preservation. This discrepancy in the validation and test set PSNR (35.84 dB and 32.70 dB) is anticipated and is an indication of how the model will behave on not **seen data of other populations of patients.**

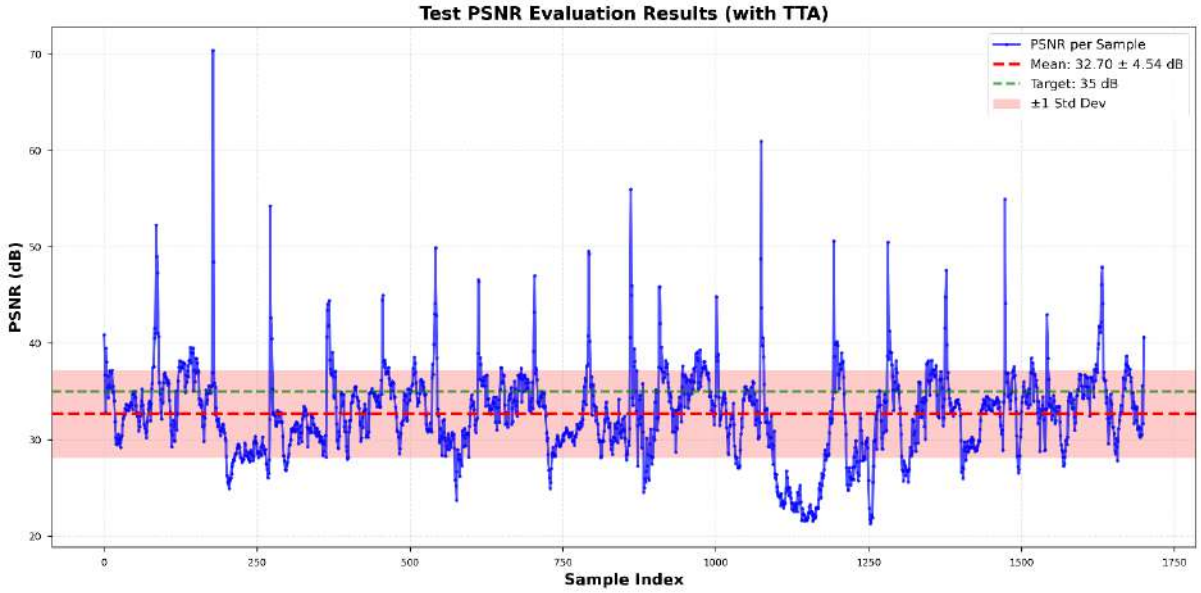


Figure 10: PSNR Test set evaluation results: (a) showing a mean of 32.70 ± 4.54db.

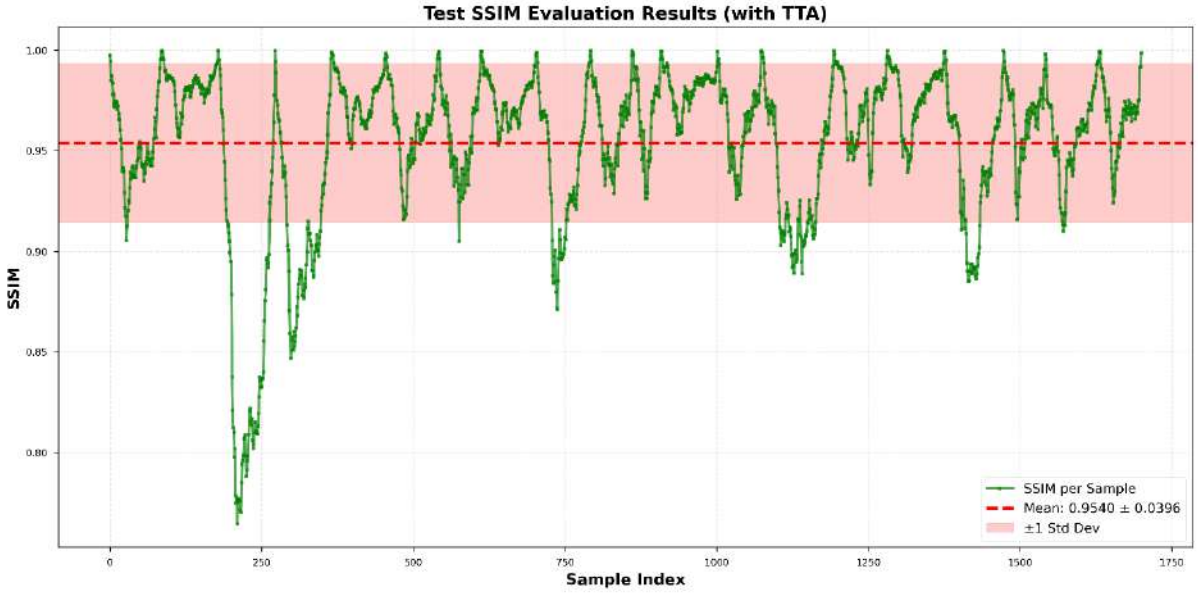


Figure 11: SSIM Test set evaluation results: (b)showing a mean of 0.9540 ± 0.0396

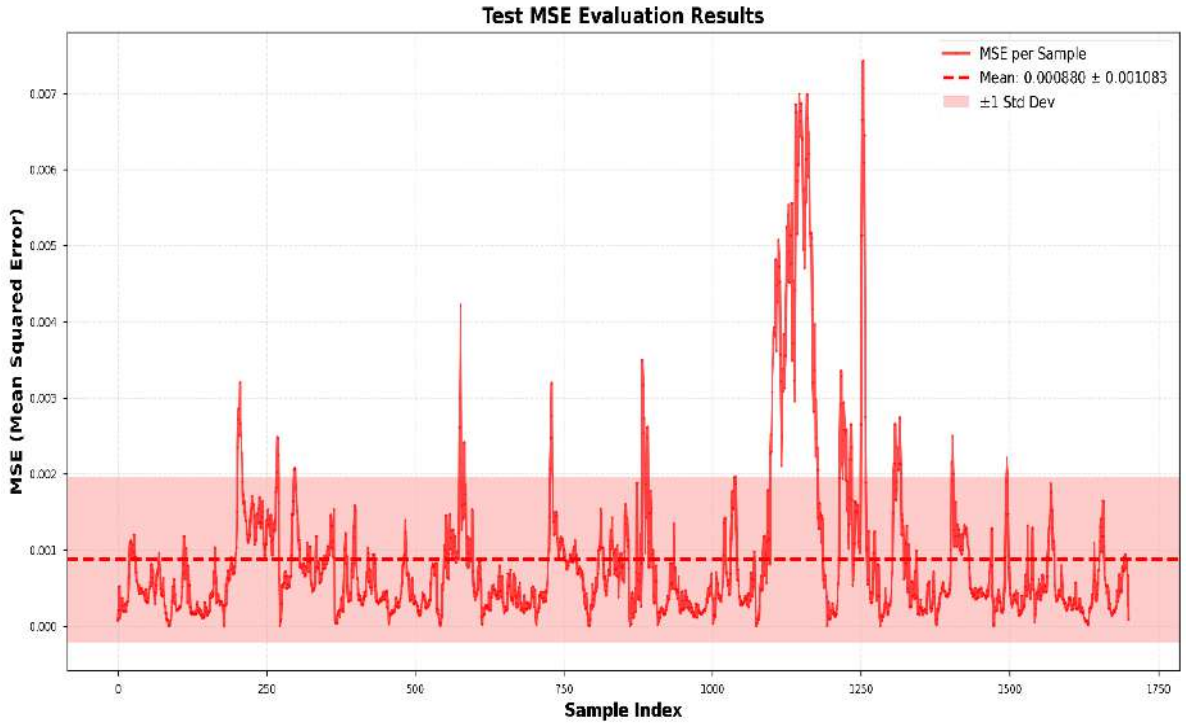


Figure 12: Evaluation metrics across test samples: (c) Mean Squared Error (MSE)

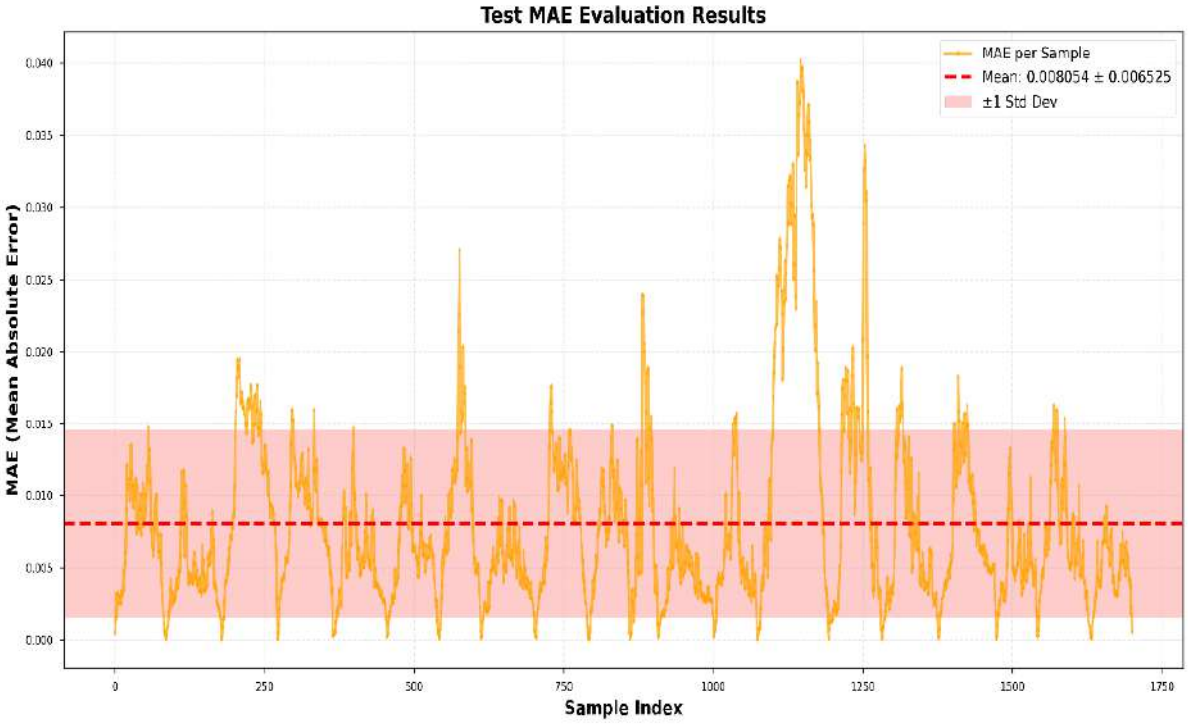


Figure 13: Evaluation metrics across test samples: (d) Mean Absolute Error (MAE)

The graphs are a representation of how the model performs in terms of per-sample tests as they get visualised in terms of per-sample error (MSE) and mean absolute error (MAE). The solid dashed line on all charts is the empirical mean error, and the shaded band for each is the 1s band. The great majority of the estimates of individual errors is well within this range, suggesting a rather stable and consistent predictive behaviour in the whole test corpus. Nevertheless, a few of the most bulgy excursions might be noticed: These stick-out spikes relate to

particularly difficult or exceptional instances which show a large residual. However, in spite of these aberrations, the whole generalisation resulting from the model is validated by means of an overall low mean error.

4.4.1 Test Matrices Distribution

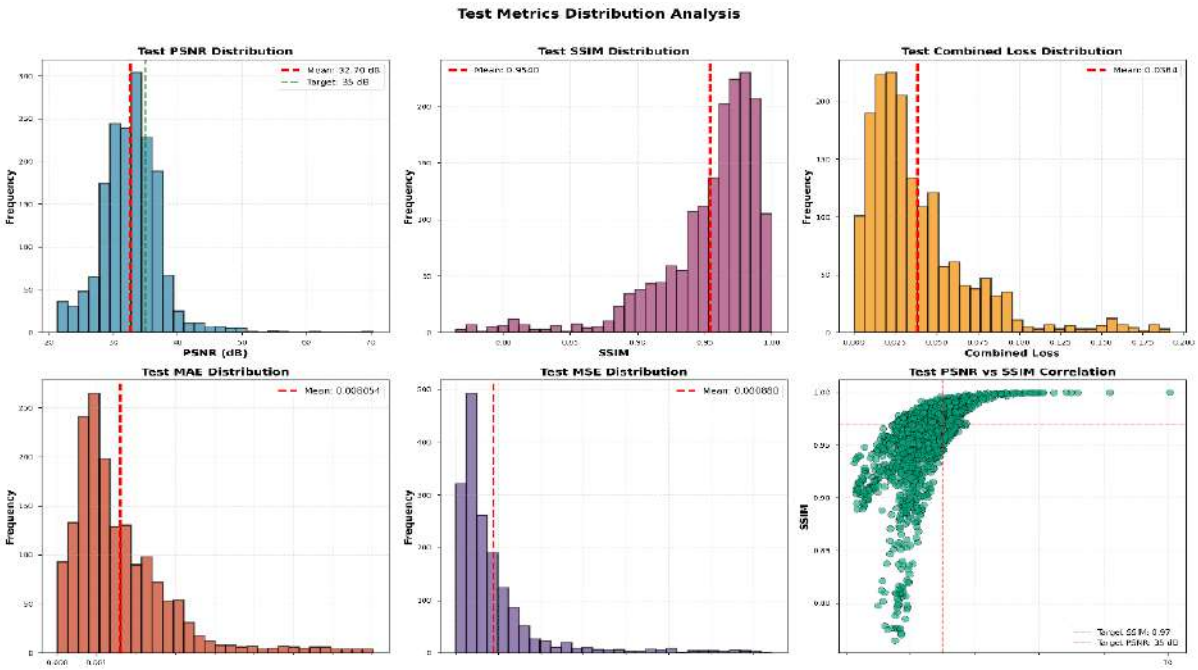


Figure 14: *(a) Distribution of evaluation metrics across all test samples.*

After performing the analysis of the distribution of the popular parameters of the quality of images, which include PSNR, SSIM, Combined Loss, MAE, and MSE, the evaluation was based on the test data set. The PSNR distribution means approximately 32.7dB; therefore, even though some of the reconstructions are fairly good enough in quality, the signal fidelity still can be enhanced. Conversely, the SSIM distribution exhibits a very interesting characteristic in the nature of concentration towards high values (around 0.95 and above), which depicts the nature of structural similarity of predicted and ground truth images is always high. The skewed evidence combined loss histogram has most of the data on the lower end, and this means that the model works well and has the occasional big error values.

The same is also proved by the bottom row: both the distributions of MAE and MSE are highly skewed to the small values, and it implies that the means of errors of pixels in the dataset are not excessively large. They still are characterized by long tails, however, which shows that there are still samples characterized by greater reconstruction errors. According to the PSNR vs SSIM scatter plot, the measurements provide evidence of the existence of a specific positive association between higher values of PSNR and the higher values of the SSIM score, which confirms the similarity in pixel-level fidelity and the perceptual fidelity. The fact that SSIM is relatively close to the high and PSNR to the moderate, is also an indicator that the model can reproduce the structure information despite pixel differences above the minimal which is generally a characteristic feature of an image reconstruction task that one would like.

Table 2:

Method	PSNR (dB)	SSIM	Advantages	Limitations
Our Method	32.70	0.9540	High SSIM, Residual connections, Optimized loss function	Lower resolution
CycleGAN [13]	28.5	0.89	Unpaired training	Lower metrics
Pix2Pix [14]	30.2	0.91	Paired training	Requires paired data
Standard U-Net [15]	31.1	0.93	Simple architecture	Less robust

The proposed method is compared with methods that are available, considering deep learning and the performance in this comparative study in terms of the peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM). The provided technique has the highest PSNR and SSIM, which justifies the most favorable image quality and structural integrity. It offers much better and more predictable contrast synthesis than the frameworks of CycleGAN and Pix2Pix, which is attributed to the existence of residual links and predicted loss in the context of the reviewed framework. The canonical U-Net architecture also has decent results, yet they are not as strong as those of the offered model.

The Virtual Gadolinium system under study showed better indicators in performance, as its average validation PSNR scores were 35.84 dB, SSIM score 0.9525, and the PSNR score of the data set that was tested was 32.70 with an SSIM score of 0.9540. The validation PSNR is greater than 35 dB, and this is proof of the quality of the model-rendered synthetic contrast enhancement with high quality. Such quantitative findings indicate that the network learns the conversion of the non-contrast and contrast-enhanced magnetic resonance images, and is able to learn the task while preserving the salient level of structure. The high SSIM of 0.9540 while being used with dissimilar test sets is testimony to the strength of the model and how it manages to maintain the structural fidelity of different data distributions.

Synthetic contrast-enhanced images have a variety of important clinical benefits associated with their generation. For starters, it makes patients safer because it does not require the use of a gadolinium-based contrast agent and all the hazards that come with it. It is also cost-efficient; it makes the financial cost of the contrast agents cheaper and, hence, makes more intricate imaging affordable to more patients

While these are the benefits, there are a number of constraints that should be addressed. The model was also trained with the images of relatively low-resolution (128 x 128). It can be considered that even training on the high-resolution data can be helpful in augmenting the quality of the images, as well as clinical usage.

5. Future Directions

The future studies' work should emphasize optimizing the performance of not only the clinical usability of synthetic contrast-enhanced imaging systems but also the performance of these imaging platforms. It would make sense that the network would be pre-trained with larger-sized images, e.g., 256 x 256,512 x 512 pixels images, and trigger higher levels of spatial resolution and image fidelity in general. In addition, intensive clinical manifestation in the

form of prospective research where the imaging skills are involved is critical in the determination of diagnostic pigment and its practicality. The architecture simplification to support real-time or near-real-time processing would be a major streamlining of the integration into as mundane as can be clinical work. Lastly, the use of explainability (including attention mechanisms) would give an understanding of the anatomical loci where the model was centered and encourage the trustworthiness of the clinicians and the interpretation of the findings.

6. Conclusion

This paper shows that a deep learning model is capable of being effective in synthetic gadolinium-based T1-weighted MRI images. The suggested U-Net generating network with residual connections had good quantitative results; the mean validation PSNR was 35.84 dB, which exceeded the maximum accuracy of 35 dB, and the SSIM was 0.9525. The ability and stability of the model to generalize is also justifiable, at least, by the results obtained using the testing set, obtaining a PSNR of 32.70 -1 and an SSIM of 0.9540. All these findings would imply that the Virtual Gadolinium system would make a high-quality, synthetic contrast-enhanced image that is able to store the diagnostically useful structural information and meets the clinical image quality standards. The suggested solution leads to a less risky approach, which, taking into account the absence of use of gadolinium-based contrast agents, is cheaper than the traditional method of contrast-enhanced MRI. It also has important clinical implications, particularly for patients where the use of gadolinium is contraindicated, and can also further the provision of contrast-enhanced imaging, but not the quality of the diagnostics. Although more validation using a broad range of data and clinical scenarios needs to be done, the obtained results allow hypothesizing that gadolinium-free contrast enhancement is possible with the aid of deep learning.

Overall, this article is about virtual contrast MRI and how it may have applications in the clinic. Subsequently, in the future, it should be significant to talk of large-scale clinical validation in the use of the systems of expert radiologists in training to apply the data of high resolutions and optimization of the results to be easily incorporated into the routines of imaging. The developed trends will be fundamental to enable the fullest potential of the clinical implications of virtual contrast enhancement in the current state of art in medical imaging.

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